

GRIDSHIELD AI — AUTONOMOUS PURPLE SOC ARCHITECTURE

E.ON Innovation Challenge 2026 // Distributed Energy Resource (DER) Grid Security

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Core Mission: Protecting 9.6M Customer DER Assets & High-Voltage Substations using GraphSAGE GNN Blast Radius Correlation and 1.14ms Edge TinyML Defense.

- **Live Application Demo:** <https://e-on-project.vercel.app/>
- **GitHub Repository:** <https://github.com/Kabirroy12345/E.ON-Project> (Branch: v2-global-edition)

THE PROBLEM: THE UNMANAGED DER GRID THREAT

Massive Edge Expansion Creates Critical Vulnerabilities for DSOs

- **Explosive Edge Growth:** Millions of residential solar inverters, EV chargers, and battery storages connecting directly to DSO distribution grids.
- **High-Voltage Cascades:** Unmanaged IoT endpoints create unmonitored backfeed pathways capable of compromising substation transformers.
- **Privacy & SIEM Cost Bottleneck:** Centralized cloud log collection violates residential GDPR privacy and costs over €4.50/device/year.

THE SOLUTION: DUAL-ENGINE PURPLE TEAM ARCHITECTURE

Autonomous Offensive Simulation & Real-Time Graph Neural Network Defense

- **Offensive Red Team Engine:** Autonomous AI executing 50+ MITRE ATT&CK; adversary vectors against SCADA, Modbus, and OCPP protocols.
- **Defensive Blue Team Engine:** Real-time PyTorch GraphSAGE 2-hop neighborhood sampling evaluating threat propagation across subgraphs in **<140ms** on GPU.
- **Automated Isolation:** Air-gapped zero-trust containment policy isolating infected gateways within milliseconds.

SILICON-LEVEL PRIVACY: 1.14ms EDGE TINYML AGENT

Zero Telemetry Leakage on ARM Cortex-M4 Microcontrollers

- **On-Device Inference:** Compiled C++ neural weights with TFLite Micro & CMSIS-NN executing on standard smart meter gateways.
- **Ultra-Low Overhead:** 1.14ms target inference latency with **<800KB RAM footprint** allowing 800+ evaluations/sec.
- **100% GDPR Compliant:** Raw customer usage data never leaves the device — only anomalous risk embeddings trigger cloud alerts.

CLUSTER 5 CUSTOMER ADOPTION FRAMEWORK

Securing 90%+ Voluntary Customer Enrollment

- **1. Dynamic Grid Tariff Discounts:** Reduced electricity rates for enrolled smart solar & battery assets.
- **2. Up to 25% Insurance Rebates:** Partnered cyber & grid risk insurance premium reductions.
- **3. Free Automated Firmware Updates:** Over-the-air Zero-Day vulnerability patching.
- **4. Priority Restoral SLAs:** Guaranteed top-tier grid restoral priority during outages.

GLOBAL MARKET SCALABILITY & COMPLIANCE

Extending Beyond EU NIS2 to Global Energy Grids

- **EU NIS2:** Full alignment with Article 21 mandatory technical measures and incident reporting.
- **US NERC-CIP & FERC 2222:** Compliant with NERC-CIP-003/005/012 cyber security standards.
- **Australia AEMO & ASEAN:** Optimized for rooftop solar heavy feeders and air-gapped island microgrids.

BUSINESS IMPACT & ITEMIZED TCO BASELINE

€18.4M Annual Savings & 5.6x ROI for a 5M Asset DSO

- **Cost Baseline:** Ingestion overhead reduced from €4.50 to **€0.80 / device / year** at scale.
- **99.2% Edge Filtering:** Local microcontroller telemetry processing eliminates 99.2% of raw log storage costs.
- **Financial ROI:** 5.6x return on investment with payback period under 8 months.